attaching carriages and ferrying people. [11] This line, because of the runs on it, indisputably remains the documented ancestor of the Indian Railways.

The experiments were first conducted at the lower Annicut (an irrigation weir) across the Coleroon river, close to Trichinopoly, and later at the Porto Novo Iron Works, once located on the Coromandel coast. The trials were documented a year later (January 1839) by Captain John Thomas Smith in an official and private compilation titled *Reports, Correspondence and Original Papers on Various Professional Subjects Connected with the Duties of Corps of Engineers of Madras Presidency* .

The object of the proposal, writes Captain Smith in the preface of the compilation, was to provide a record of the experience of its several members for the benefit of all and 'keep up a more lively interest in duties'.

In a chapter titled 'Notes of Some Experiments Made by Captain A. Cotton, upon Mr Avery's Steam Engine', Captain Smith says he managed to lay his hands on a few of the experimentation notes by A.T. Cotton and published them for wider use. Some excerpts below might be useful to the reader.

> A great variety of highly interesting experiments have been lately made by Captain A. Cotton, on the subject of the efficiency of Mr Avery's Steam Engine, and its fitness for locomotive, and other purposes in India.

He notes the technicalities of the new find and its method of usage.

THE FIRST EXPERIMENTS OF THE 'NEW MECHANICAL AGENT'

The first experiments which were made on this engine were tried at the lower Annicut (irrigation weir) across the Coleroon, [12] about 35 miles below Trichinopoly. [13]

> A small machine was here made for pounding of magnesia, and by it a hammer weighing 30 lbs was tilted by the usual contrivance of a set of wipers fixed on the circumference of a cylinder, which was made to turn by a strap passing around a large band wheel fixed on it, and over a pulley on the same axle with arms. The boiler which supplied this engine was ascertained by experiments, to evaporate about ¾ of a cubic foot of water per hour, which was made use of in the condition of steam of 30 lbs, effective pressure… [14]

But this experiment, conducted on a small scale, was not enough to generalize the results. Larger experiments continued being administered at the Porto Novo Iron Works.